

# Spearman's rank correlation coefficient



1

- a Pearson's correlation coefficient because there is a clear upward trend.
- b Spearman's rank coefficient because the trend is not linear.

2

a

i

|             |   |     |      |      |      |     |      |
|-------------|---|-----|------|------|------|-----|------|
| $x$         | 3 | 4   | 5    | 6    | 7    | 8   | 9    |
| Rank of $x$ | 1 | 2   | 3    | 4    | 5    | 6   | 7    |
| $y$         | 7 | 7.4 | 7.88 | 7.64 | 7.72 | 8.2 | 7.32 |
| Rank of $y$ | 1 | 3   | 6    | 4    | 5    | 7   | 2    |

ii  $r_s = 0.36$

iii Weak, positive correlation

b

i

|             |     |     |     |     |     |     |
|-------------|-----|-----|-----|-----|-----|-----|
| $x$         | 2   | 6   | 7   | 14  | 17  | 22  |
| Rank of $x$ | 1   | 2   | 3   | 4   | 5   | 6   |
| $y$         | 2.0 | 2.4 | 2.0 | 0.6 | 0.9 | 0.2 |
| Rank of $y$ | 4.5 | 6   | 4.5 | 2   | 3   | 1   |

ii  $r_s = -0.84$

iii Strong, negative correlation

c

i

|             |    |      |      |       |       |       |       |
|-------------|----|------|------|-------|-------|-------|-------|
| $x$         | 1  | 4    | 7    | 10    | 13    | 16    | 19    |
| Rank of $x$ | 1  | 2    | 3    | 4     | 5     | 6     | 7     |
| $y$         | -1 | -1.7 | -1.7 | -1.84 | -1.91 | -1.98 | -2.05 |
| Rank of $y$ | 7  | 5.5  | 5.5  | 4     | 3     | 2     | 1     |

ii  $r_s = -0.99$

iii Strong, negative correlation

d  
i



|             |     |     |     |     |     |     |
|-------------|-----|-----|-----|-----|-----|-----|
| $x$         | 4   | 6   | 8   | 13  | 17  | 21  |
| Rank of $x$ | 1   | 2   | 3   | 4   | 5   | 6   |
| $y$         | 0.4 | 0.9 | 0.6 | 1.7 | 2.4 | 1.7 |
| Rank of $y$ | 1   | 3   | 2   | 4.5 | 6   | 4.5 |

ii  $r_s = 0.84$

iii Strong, positive correlation

3

a  $r_p = 0.958$

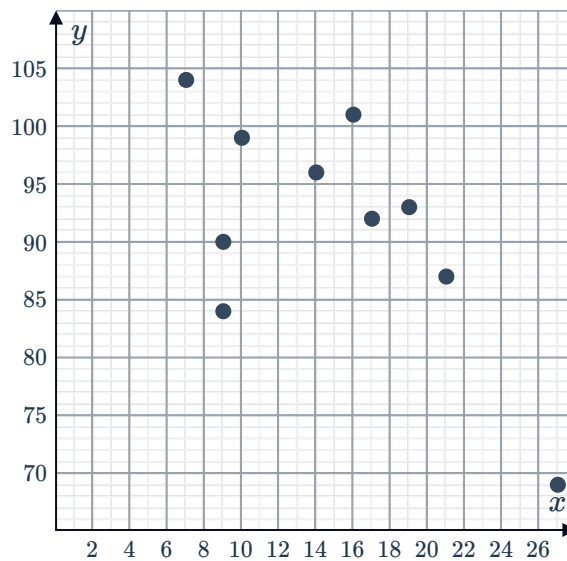
b  $r_s = 1.00$

c Very strong positive correlation.

## Applications

4

a



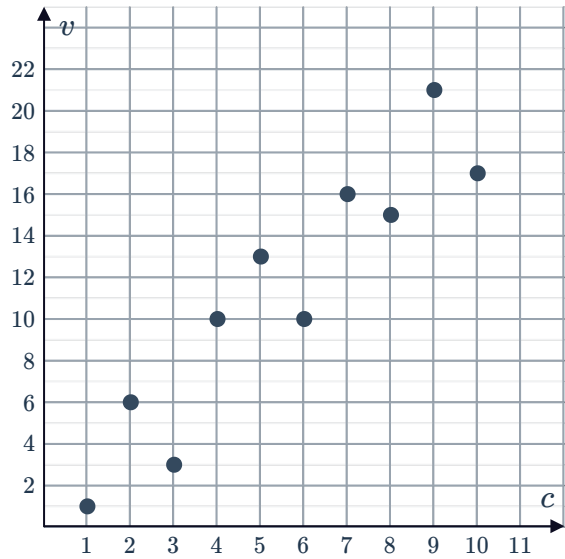
b

|                            |    |    |     |     |     |    |    |    |     |    |
|----------------------------|----|----|-----|-----|-----|----|----|----|-----|----|
| Age when first spoke, $x$  | 14 | 27 | 9   | 9   | 16  | 21 | 17 | 10 | 7   | 19 |
| Rank of $x$                | 5  | 10 | 2.5 | 2.5 | 6   | 9  | 7  | 4  | 1   | 8  |
| Aptitude test results, $y$ | 96 | 69 | 84  | 90  | 101 | 87 | 92 | 99 | 104 | 93 |
| Rank of $y$                | 7  | 1  | 2   | 4   | 9   | 3  | 5  | 8  | 10  | 6  |

c  $r_s = -0.444$

5

a



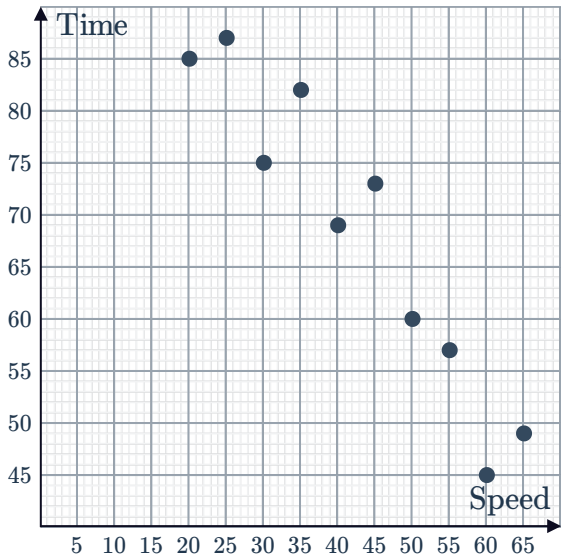
b

|              |   |   |   |     |    |     |    |    |    |    |
|--------------|---|---|---|-----|----|-----|----|----|----|----|
| Current, $c$ | 1 | 2 | 3 | 4   | 5  | 6   | 7  | 8  | 9  | 10 |
| Rank of $c$  | 1 | 2 | 3 | 4   | 5  | 6   | 7  | 8  | 9  | 10 |
| Voltage, $v$ | 1 | 6 | 3 | 10  | 13 | 10  | 16 | 15 | 21 | 17 |
| Rank of $v$  | 1 | 3 | 2 | 4.5 | 6  | 4.5 | 8  | 7  | 10 | 9  |

c  $r_s = 0.942$

6

a



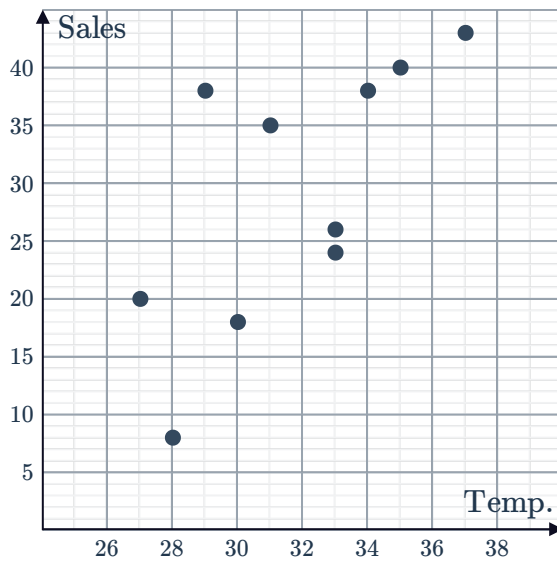
b  $r_s = -0.952$

c Strong, negative, linear correlation

7



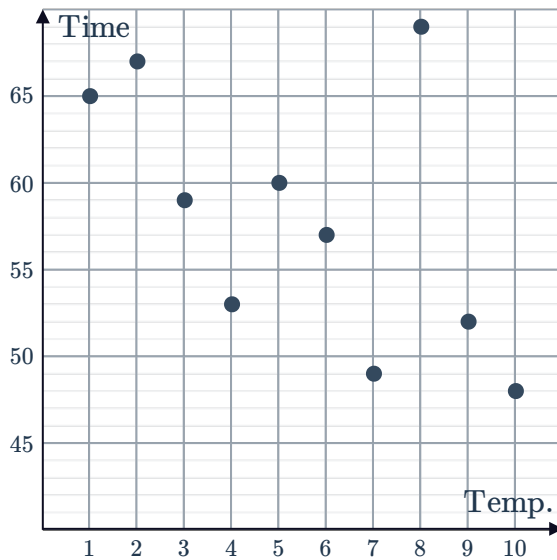
a

b  $r_s = 0.76$ 

c The sales tend to increase.

8

a



b 10

c  $r_s = -0.564$ 

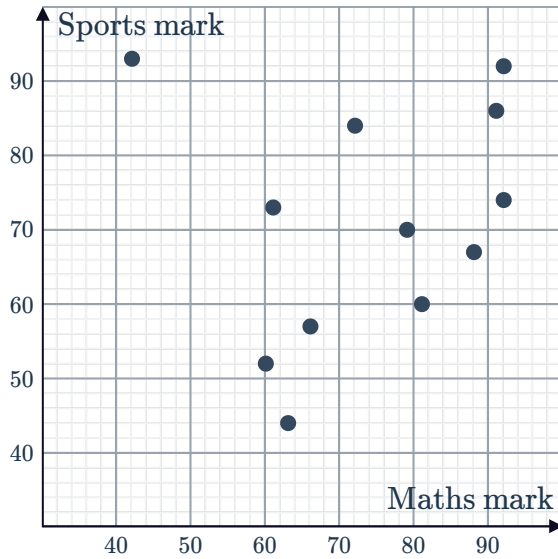
d Moderate, negative, linear correlation

e  $8^\circ\text{C}$ , 69 secondsf  $r_s = -0.883$

9



a



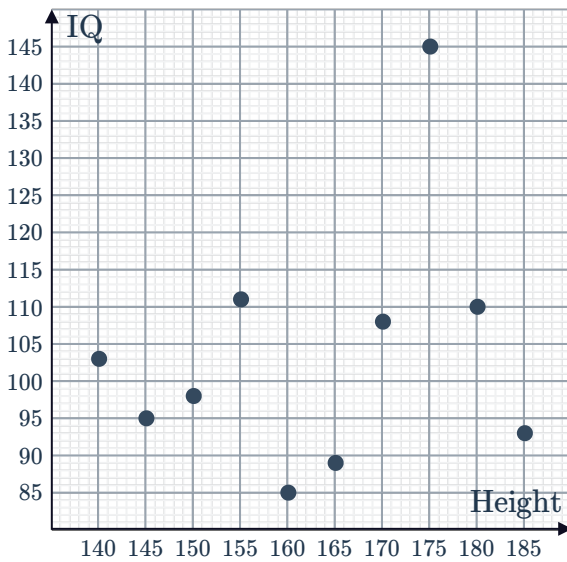
b Weak, positive, linear correlation

c Student 10's

d  $r_s = 0.670$

10

a



b Not correlated

c Height = 175 cm

d  $r_s = -0.05$